United States International University-Africa

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School of Science and Technology

Course: DSA 3050A / Advanced Business Intelligence & Data Visualization

Advanced Power BI Examination Report:

Healthcare Operations and Patient Flow Analytics

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This project demonstrates the end-to-end development of an advanced Business Intelligence solution designed to address critical operational bottlenecks within a multi-facility hospital network. Utilizing Microsoft Power BI, this report details the transformation of a raw, 100,000-row healthcare dataset into an interactive, enterprise-grade analytical dashboard. The primary objective of this solution is to provide hospital administrators and clinical directors with granular visibility into patient flow dynamics, specifically focusing on the core metrics of Average Length of Stay (LOS) and hospital readmission rates.

Part A: Data Acquisition and Understanding

1. Domain and Business Problem Being Solved

Domain: Healthcare Analytics and Clinical Operations

Business Problem: A healthcare network is seeking to understand the key drivers behind patient hospitalization durations. Administrators and clinical managers lack clear visibility into how specific patient demographics, underlying comorbidities, and admission vitals impact the overall length of stay and readmission rates across different facilities.

The objective of this Power BI solution is to develop an analytical model that enables healthcare stakeholders to:

- Analyze the average Length of Stay (lengthofstay) across different facilities and patient profiles.
- Identify which underlying comorbidities (e.g., asthma, iron deficiency, malnutrition) most heavily influence prolonged hospitalizations.
- Evaluate the correlation between initial admission lab results/vitals (e.g., glucose, BMI, pulse) and readmission counts (rcount).
- Monitor patient volume and discharge trends over time to better anticipate facility resource needs.

2. Dataset Source

Source: The data is sourced from a comprehensive "Hospital Length of Stay" dataset (available via platforms like Kaggle or Microsoft). Format: Acquired as a raw flat file, the dataset contains thousands of patient encounter records, capturing dates, facility routing, patient demographics, clinical vitals, and lab results upon admission.

3. Key Tables and Fields

The raw dataset is provided as a wide, 28-column flat file. To adhere to business intelligence best practices, this flat file will be transformed via Power Query into a relational Star Schema. The eid (Encounter ID) will serve as the primary linkage point.

- Fact_Admissions (Transactional Table)

    - Keys: eid (Encounter/Primary Key), facid (Facility Foreign Key)
    - Dates: vdate (Visit/Admission Date), discharged (Discharge Date)
    - Metrics: lengthofstay (Days), rcount (Readmission Count)
    - Attributes: secondary (Secondary Diagnosis)
- Dim_Patient_Profile (Dimension Table)
    - Fields: eid (Key), gender, and Boolean comorbidity flags including dialysisren, asthma, irondef, pneum (pneumonia), substance (substance abuse), psychologi, depress, psychothe, fibrosisand, and malnutritio.
- Dim_Clinical_Records (Dimension Table)
    - Fields: eid (Key), Vitals (bmi, pulse, respiration), and Lab Results (hemo, hematocri, neutrophils, sodium, glucose, bloodurea, creatinine).
- Dim_Facilities (Dimension Table)
    - Fields: facid (Primary Key). Note: This table will be created in Power Query by extracting unique facility IDs from the main table.

4. Suitability for Advanced Power BI Analysis

This healthcare dataset is exceptionally suited for an advanced Power BI examination for the following reasons:

- Extensive Variable Mix: With 28 distinct columns covering categorical data (gender), boolean flags (comorbidities), numerical metrics (labs, vitals, LOS), and dates, the dataset provides a rich foundation for complex data visualization and interactive drill-downs.
- Dimensional Modeling Potential: The wide flat file practically mandates the use of Power Query to unpivot, duplicate, and normalize the data into a formal Star Schema, perfectly demonstrating advanced data transformation skills.
- Advanced DAX Opportunities: The presence of two distinct date columns (vdate and discharged) requires the use of disconnected date tables and functions like USERELATIONSHIP to calculate active hospital capacity. Furthermore, the extensive clinical metrics allow for advanced statistical DAX measures (e.g., calculating average lengthofstay strictly for patients where malnutritio = True).
- Operational KPIs: It shifts the analytical focus from standard financial sums to operational averages, conditional formatting based on healthy clinical ranges (e.g., flagging abnormal glucose or pulse), and resource tracking.

AAYUSH CHOUDHURY · UPDATED 4 YEARS AGO

12 Code Download

Hospital Length of Stay Dataset Microsoft

100k samples with health indicator features useful for Length of Stay Prediction

Data Card Code (2) Discussion (2) Suggestions (0)

About Dataset

This dataset has 100k data points on patients admitted into hospital, indicators of their health condition and how long they were admitted in the hospital. It been taken from this website which contains the full data description: https://microsoft.github.io/r-server-hospital-length-of-stay/input_data.html

The dataset is useful for predicting length of stay within hospital, it has been open sourced by Microsoft and I have been using it for experimentation, thought I'd share it with the rest of the community.

Follows: https://opensource.microsoft.com/codeofconduct/

The features I found that were most useful were readmission count and an engineered feature (total number of issues) which sums the binary columns.

Usability

10.00

License

Other (specified in description)

Expected update frequency

Never

Tags

Health Tabular Healthcare

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	eid	vdate	rcount	gender	dialysisren	asthma	irondef	pneum	substance	psychologi	depress	psychothe	fibrosisanc	malnutritic	hemo	hematocri	neutrophil	sodium	glucose	bloodurea	creatinine	bmi	pulse
1		########	0	F	0	0	0	0	0	0	0	0	0	0	0	11.5	14.2	140.3611	192.4769	12	1.390722	30.43242	96
2		########	5+	F	0	0	0	0	0	0	0	0	0	0	0	9	4.1	136.7317	94.07851	8	0.943164	28.46052	61
3		########	1	F	0	0	0	0	0	0	0	0	0	0	0	8.4	8.9	133.0585	130.5305	12	1.06575	28.84381	64
4		8/9/2012	0	F	0	0	0	0	0	0	0	0	0	0	0	11.9	9.4	138.994	163.377	12	0.906862	27.95901	76
5		########	0	F	0	0	0	1	0	1	0	0	0	0	0	9.1	9.05	138.6348	94.88665	11.5	1.242854	30.25893	67
6		########	3	M	0	0	0	0	0	0	0	0	0	0	0	14.8	17.8	136.5931	120.2653	11	0.884663	27.87594	83
7		########	4	F	0	0	0	0	0	0	0	0	0	0	0	9.4	8.5	137.1513	124.3357	6	0.923084	29.92882	68
8		6/4/2012	0	F	0	0	0	0	0	1	0	0	0	0	0	11.25	7.15	136.0697	152.2659	11	0.768106	32.99917	63
9		7/5/2012	0	F	0	0	0	0	1	0	0	0	0	0	0	11.9	9.4	135.9803	100.2832	12	1.178786	28.80268	69
10		9/1/2012	0	F	0	0	0	0	0	0	0	0	0	0	0	10.7	8.5	136.8934	167.0661	10	1.585072	32.50373	65
11		########	0	F	0	0	0	0	0	0	0	0	0	0	0	12.2	9.1	140.0608	121.8991	7	1.254258	30.32516	73
12		########	4	M	0	0	0	0	0	0	0	0	0	0	0	12.6	11	137.8159	147.1848	16	1.264943	27.30172	63
13		7/8/2012	0	F	0	1	0	0	0	0	0	0	0	0	0	11.4	17.7	137.0812	127.049	17	0.98113	31.35743	82
14		########	0	F	0	0	0	0	1	0	0	0	0	0	1	8.1	16.9	134.3444	167.7208	12	1.02266	29.6929	73
15		########	1	F	0	0	0	0	0	0	0	0	0	0	0	11.9	9.4	136.0829	95.34306	12	1.05082	28.5778	69
16		########	1	F	0	0	0	0	0	0	0	0	0	1	0	11.9	9.4	145.7156	146.4891	16	1.239281	30.63941	68
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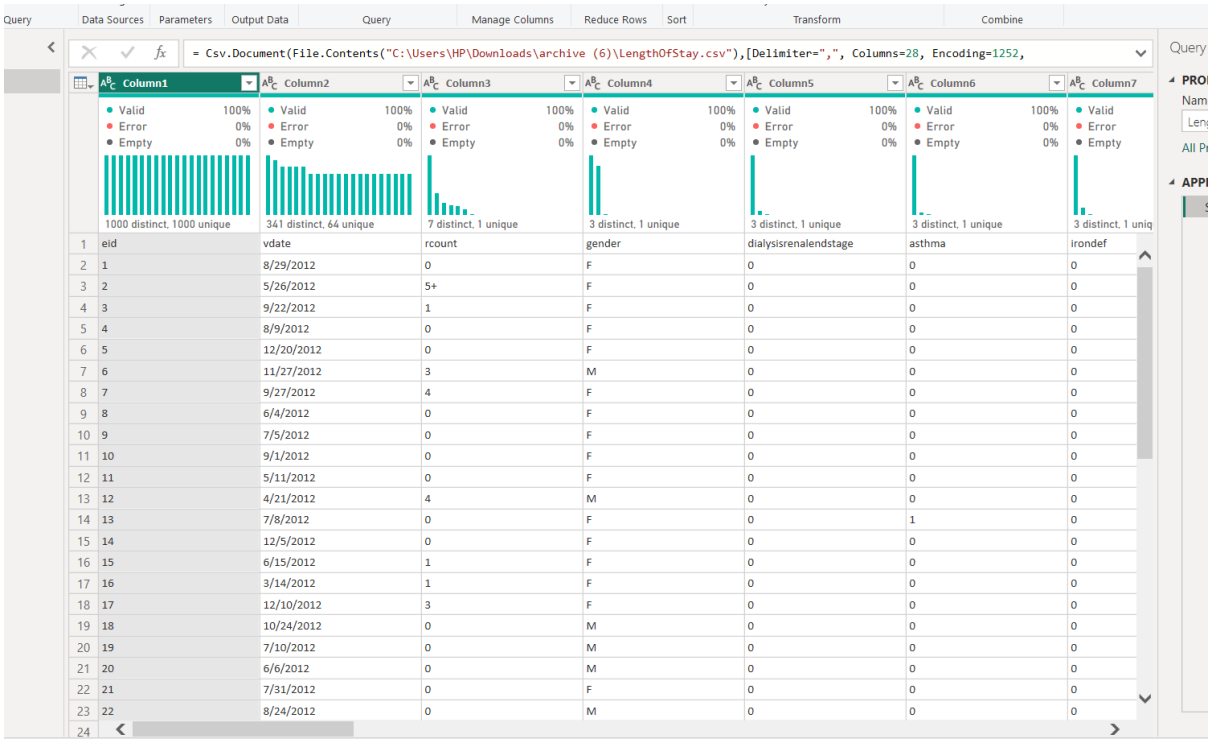
18		########	0	M	0	0	0	0	0	0	0	0	0	0	0	14.9	10.5	136.0839	64.11917	9	1.188563	31.10382	72
19		########	0	M	0	0	0	0	1	0	0	0	0	0	0	10.1	9.1	133.1922	109.3141	7	0.798395	28.98102	63
20		6/6/2012	0	M	0	0	0	0	0	0	0	0	0	0	0	13.4	16.6	137.1202	183.5635	11	1.051098	31.82269	72
21		########	0	F	0	0	0	0	0	0	0	0	0	0	0	7.7	2.8	133.4024	154.7973	10.5	1.054753	30.77001	75
22		########	0	M	0	0	0	0	0	0	0	0	0	0	0	12.2	7.4	137.4194	161.8215	12	1.332474	28.93112	77
23		########	3	M	0	0	0	0	0	0	0	0	0	1	0	13.2	6.3	137.3712	147.092	28	1.065005	28.34169	70
24		########	1	M	0	0	1	0	0	0	0	0	0	1	1	10	14.3	139.2565	122.2878	35	1.071094	29.70451	101
25		########	0	F	1	0	0	1	1	0	0	0	0	0	1	10.8	10.2	135.802	150.3622	104	1.248269	25.91784	62
26		7/8/2012	1	M	0	0	0	0	0	1	0	1	0	0	0	11.1	10.3	135.2008	148.9015	35	0.967228	31.25558	81
27		########	2	F	0	0	0	0	0	1	0	0	0	0	0	12.2	10.5	137.8581	159.2252	7	0.913336	35.48215	88

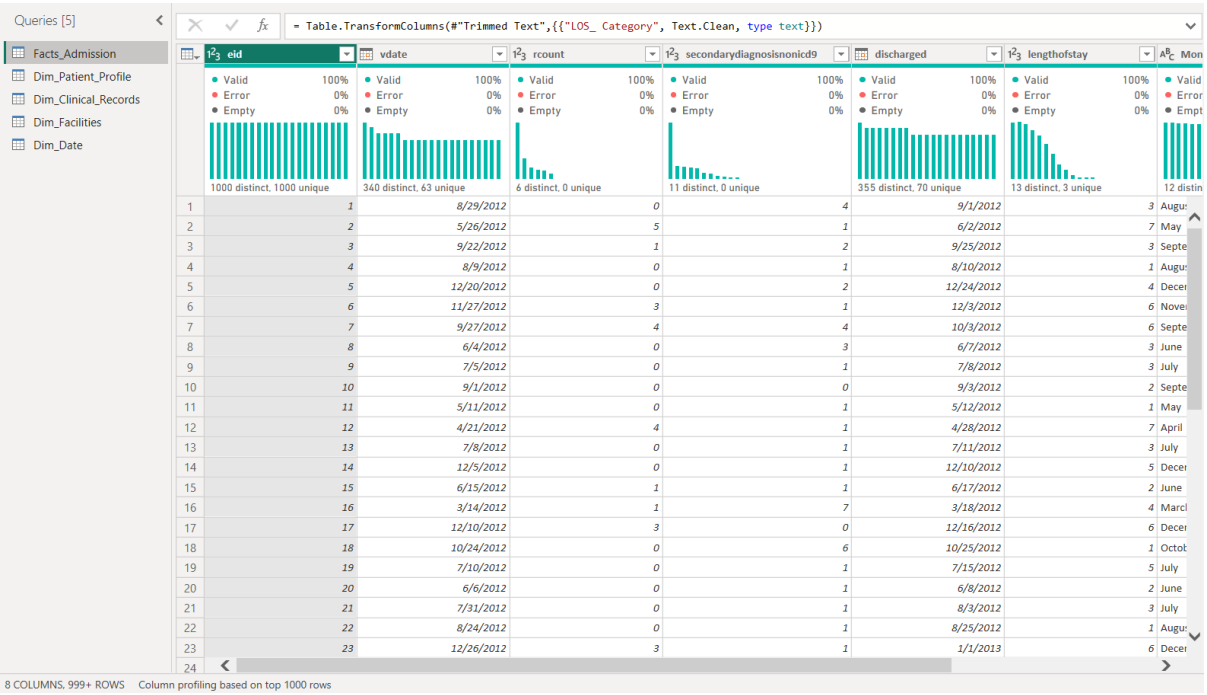
Part B: Cleaning & Transformation.

Before



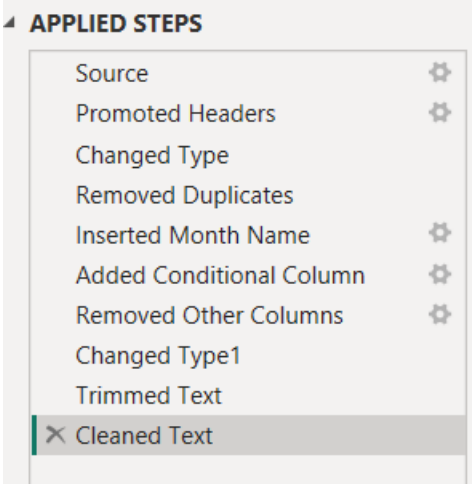
	eid	vdate	rcount	gender	dialysisrenalendstage	asthma	irondef
2	1	8/29/2012	0	F	0	0	0
3	2	5/26/2012	5+	F	0	0	0
4	3	9/22/2012	1	F	0	0	0
5	4	8/9/2012	0	F	0	0	0
6	5	12/20/2012	0	F	0	0	0
7	6	11/27/2012	3	M	0	0	0
8	7	9/27/2012	4	F	0	0	0
9	8	6/4/2012	0	F	0	0	0
10	9	7/5/2012	0	F	0	0	0
11	10	9/1/2012	0	F	0	0	0
12	11	5/11/2012	0	F	0	0	0
13	12	4/21/2012	4	M	0	0	0
14	13	7/8/2012	0	F	0	1	0
15	14	12/5/2012	0	F	0	0	0
16	15	6/15/2012	1	F	0	0	0
17	16	3/14/2012	1	F	0	0	0
18	17	12/10/2012	3	F	0	0	0
19	18	10/24/2012	0	M	0	0	0
20	19	7/10/2012	0	M	0	0	0
21	20	6/6/2012	0	M	0	0	0
22	21	7/31/2012	0	F	0	0	0
23	22	8/24/2012	0	M	0	0	0

After



	eid	vdate	rcount	secondarydiagnosisnonicd9	discharged	lengthofstay	Mon
1	1	8/29/2012	0	4	9/1/2012	3	Augu
2	2	5/26/2012	5	1	6/2/2012	7	May
3	3	9/22/2012	1	2	9/25/2012	3	Septe
4	4	8/9/2012	0	1	8/10/2012	1	Augu
5	5	12/20/2012	0	2	12/24/2012	4	Decer
6	6	11/27/2012	3	1	12/3/2012	6	Nove
7	7	9/27/2012	4	4	10/3/2012	6	Septe
8	8	6/4/2012	0	3	6/7/2012	3	June
9	9	7/5/2012	0	1	7/8/2012	3	July
10	10	9/1/2012	0	0	9/3/2012	2	Septe
11	11	5/11/2012	0	1	5/12/2012	1	May
12	12	4/21/2012	4	1	4/28/2012	7	April
13	13	7/8/2012	0	1	7/11/2012	3	July
14	14	12/5/2012	0	1	12/10/2012	5	Decer
15	15	6/15/2012	1	1	6/17/2012	2	June
16	16	3/14/2012	1	7	3/18/2012	4	March
17	17	12/10/2012	3	0	12/16/2012	6	Decer
18	18	10/24/2012	0	6	10/25/2012	1	Octob
19	19	7/10/2012	0	1	7/15/2012	5	July
20	20	6/6/2012	0	1	6/8/2012	2	June
21	21	7/31/2012	0	1	8/3/2012	3	July
22	22	8/24/2012	0	1	8/25/2012	1	Augu
23	23	12/26/2012	3	1	1/1/2013	6	Decer

Applied steps



Promoting Headers & Profiling Data Types: Reason: Necessary to ensure that Power BI's VertiPaq engine correctly interprets mathematical operations (e.g., averaging lengthofstay) and enables time-intelligence functions for date fields (vdate). Incorrect data types cause DAX errors.

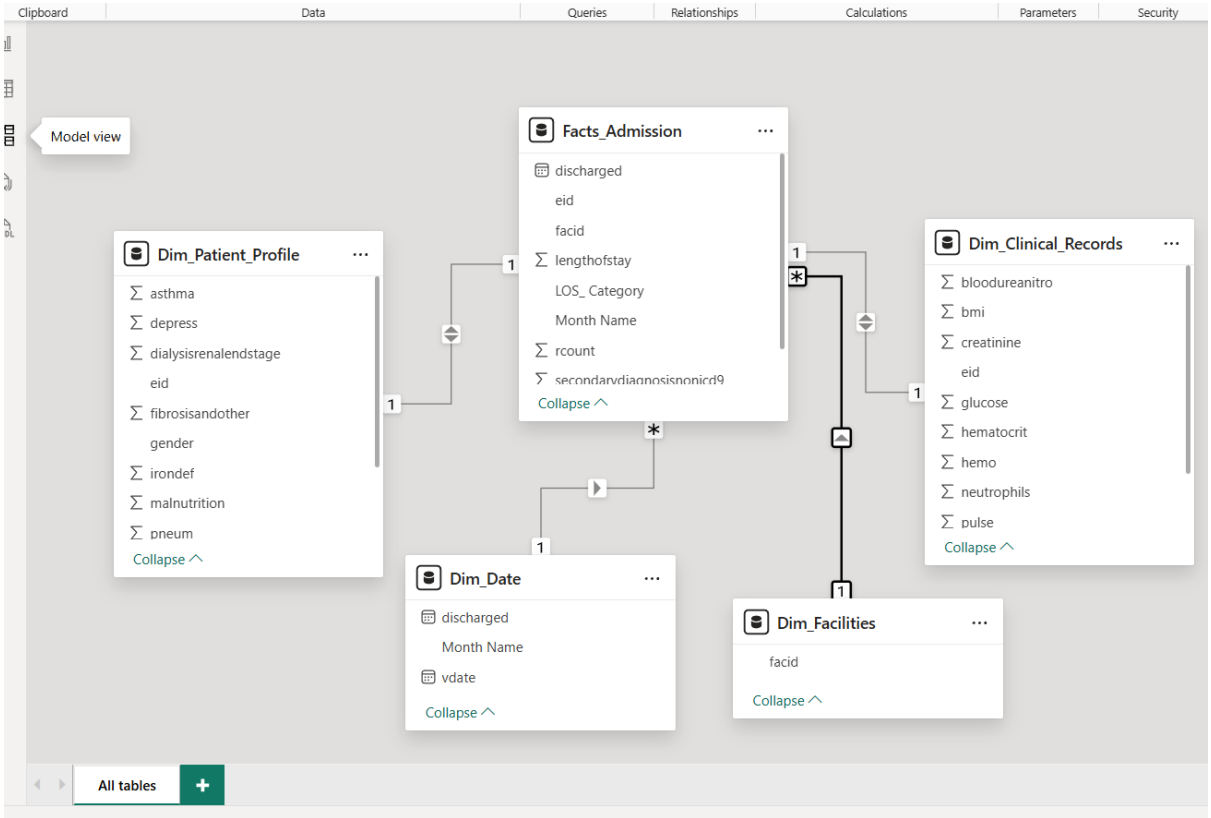
Removing Duplicates: * Reason: Crucial for establishing data integrity and creating valid Primary Keys in the dimension tables (e.g., Dim_Facilities). Without unique values, Power BI cannot establish standard "1-to-Many" relationships in the data model.

Extracting Date Parts (Month): Reason: Simplifies time-series analysis. By extracting the month and year directly in Power Query, it allows stakeholders to quickly filter admission trends without requiring the immediate development of complex DAX date formulas.

Creating Conditional Columns (LOS Category): Reason: Translates raw numbers into actionable business logic. Grouping patients into "Short Stay" or "Extended Stay" buckets allows administrators to easily filter and compare different patient cohorts.

Duplicating Queries for Dimensional Normalization: Reason: This is the most critical step for advanced data modeling. A wide 28-column flat file is inefficient; duplicating queries allows the data to be normalized into distinct tables (Patients, Facilities, Labs) to form the required Star Schema.

Part C: Data Modelling



The Fact Table: Fact_Admissions

Role & Justification: This is the quantitative core of the data model. It represents the measurable events (patient hospital encounters). I stripped out all descriptive text and isolated only the primary metrics (lengthofstay, rcount) and the essential foreign keys (eid, facid, vdate) required to link to the surrounding dimensions.

The Dimension Tables

Role & Justification: These tables contain the descriptive attributes used to filter, slice, and group the metrics in the Fact table.

Dim_Facilities: Contains facility-specific attributes.

Dim_Patient_Profile: Contains demographic data (gender) and boolean flags for underlying comorbidities (e.g., asthma, iron deficiency).

Dim_Clinical_Records: Contains the specific vitals and lab results recorded at the time of the encounter.

Dim_Date: A dedicated time-intelligence table

From Table (Dimension)	To Table (Fact)	Primary/Foreign Key	Cardinality	Cross-Filter Direction
Dim_Facilities	Fact_Admissions	facid	1 to Many (1:*)	Single
Dim_Patient_Profile	Fact_Admissions	eid	1 to 1 (1:1)	Both
Dim_Clinical_Records	Fact_Admissions	eid	1 to 1 (1:1)	Both
Dim_Date	Fact_Admissions	Date to vdate	1 to Many (1:*)	Single (Active)

Part D: DAX

i) Calculated columns

Data | Queries | Relationships | Calculations | Parameters | Security

```
Readmission_Risk_Tier =
SWITCH(
    TRUE(),
    'Facts_Admission'[rcount] = 0, "No Prior Readmission",
    'Facts_Admission'[rcount] > 0 && 'Facts_Admission'[rcount] <= 2, "Moderate Risk",
    'Facts_Admission'[rcount] > 2, "High Risk",
    "Unknown"
)
```

This column categorizes patients into risk tiers based on their historical readmission count (rcount). This allows administrators to quickly slice the dashboard to view only "High Risk" patients.

```
Is_Extended_Stay =
IF('Facts_Admission'[lengthofstay] > 5, "Yes", "No")
```

A binary flag designed to highlight hospitalizations that exceed standard care limits. This simplifies DAX percentage calculations downstream.

ii) DAX measures

```
Total Encounters = DISTINCTCOUNT('Facts_Admission'[eid])
```

Calculates the absolute baseline volume of patient visits. Using DISTINCTCOUNT on the Encounter ID ensures no accidental data duplication skews the results.

```
Average LOS = AVERAGE('Facts_Admission'[lengthofstay])
```

The core operational metric for the hospital, calculating the mean duration a patient spends in a facility.

```
Total Readmissions = SUM('Facts_Admission'[rcount])
```

Aggregates the total historical readmissions for the filtered cohort to gauge long-term care failure rates.

```
Readmission Rate % =
DIVIDE(
    CALCULATE([Total Encounters], 'Facts_Admission'[rcount] > 0),
    [Total Encounters],
    0
)
```

Calculates the percentage of total encounters that involved a patient with at least one prior readmission. Uses CALCULATE to modify the filter context in the numerator.

```
YTD Admissions = TOTALYTD([Total Encounters], 'Dim_Date'[vdate])
```

Tracks cumulative patient admissions from January 1st of the current filter year to the current date, utilizing the Dim_Date table.

```
MoM Admission Growth % =
VAR CurrentMonth = [Total Encounters]
VAR PrevMonth = CALCULATE([Total Encounters], DATEADD('Dim_Date'[vdate], -1, MONTH))
RETURN
DIVIDE(CurrentMonth - PrevMonth, PrevMonth, 0)
```

Evaluates whether hospital admissions are trending up or down compared to the previous month. It uses variables (VAR) for optimized and readable code.

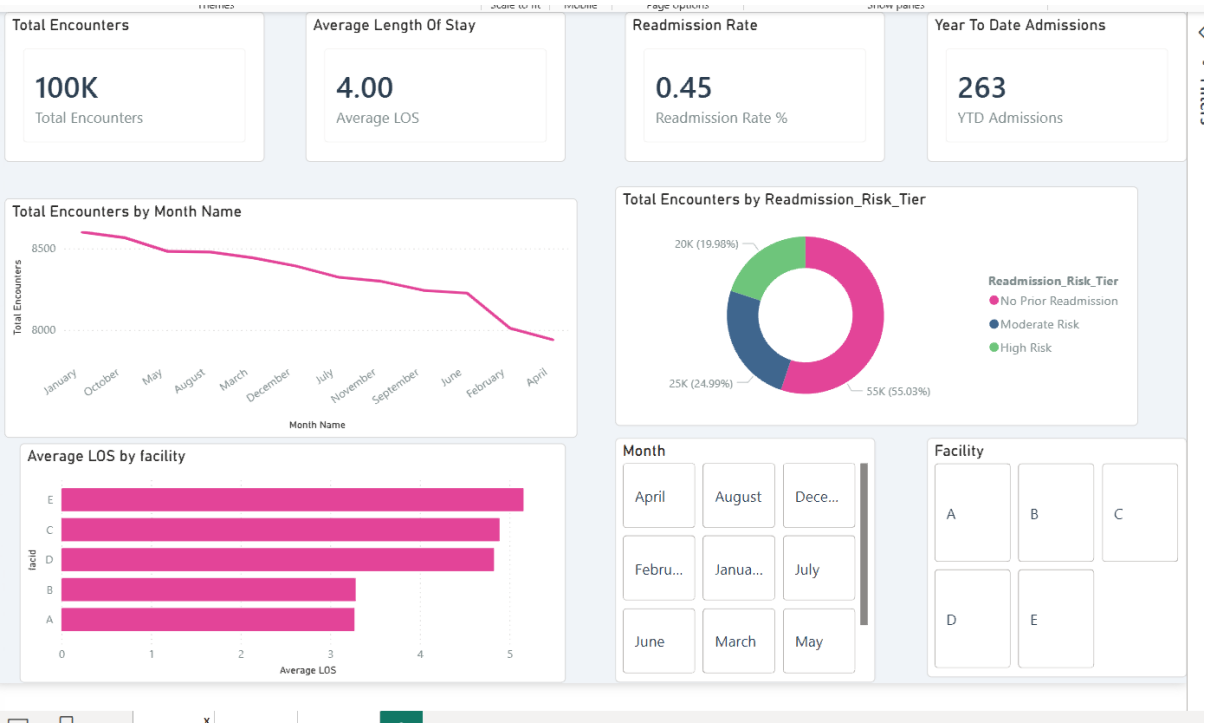
```
Facility Rank =
RANKX(
    ALL('Dim_Facilities'[facid]),
    [Total Encounters],
    ,
    DESC,
    Dense
)
```

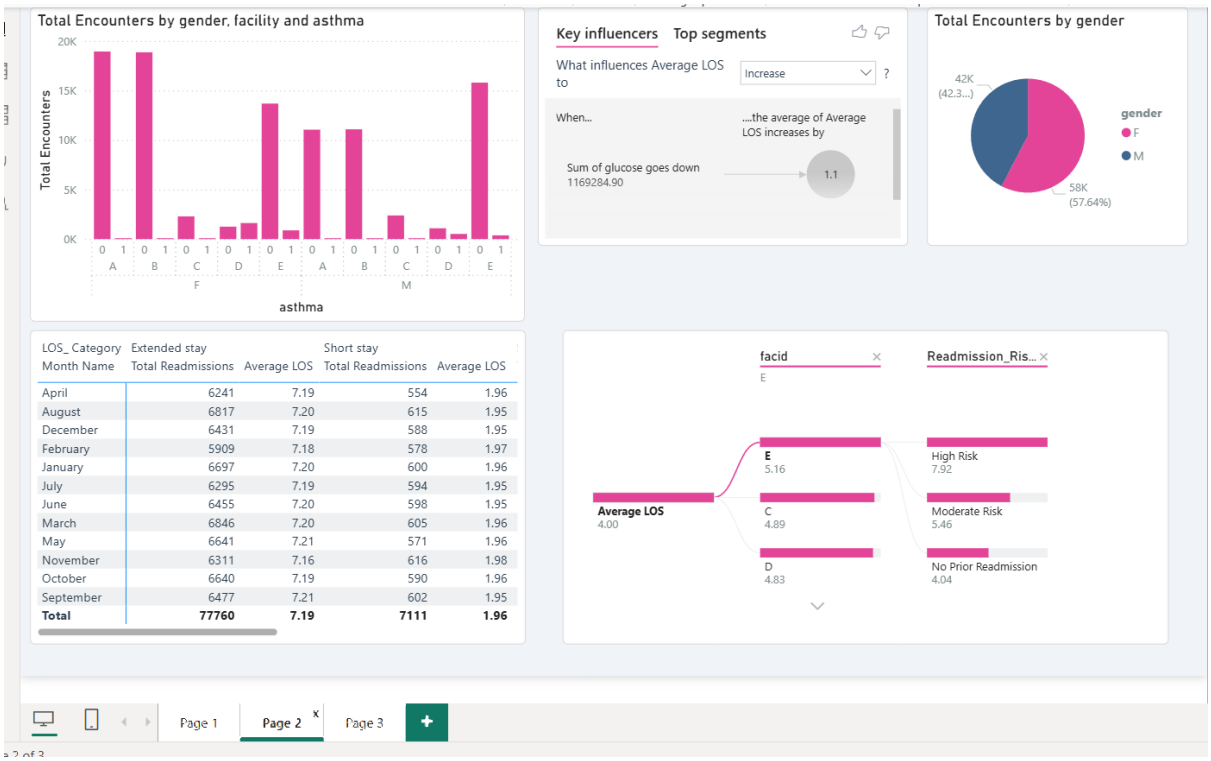
Dynamically ranks the hospital facilities from 1 to N based on their total patient encounter volume. The ALL function ensures the rank is evaluated against the entire list of facilities regardless of the current row context.

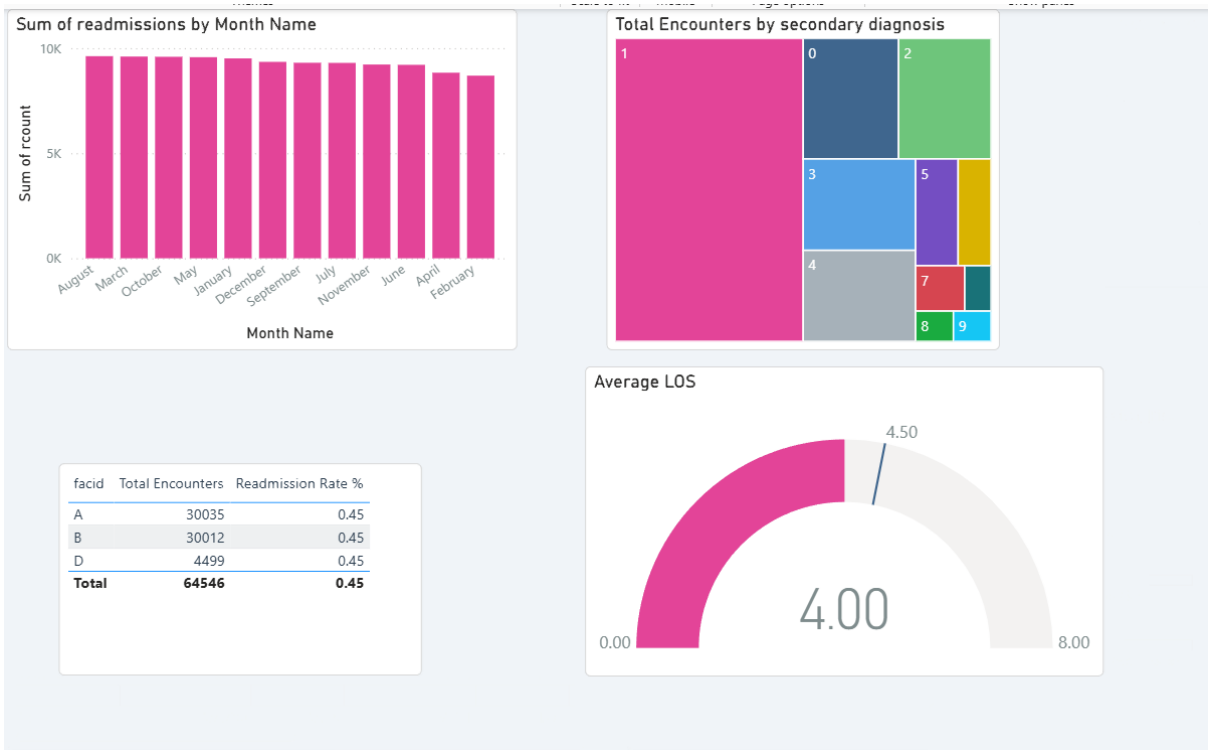
```
LOS Variance to Target = [Average LOS] - 4.5
```

I chose a strict operational target: an average Length of Stay of 4.5 days. This measure calculates how far above or below a specific facility or patient cohort is from that target.

Part E: Dashboard







Theme: A high-contrast corporate theme (white visual cards on a light-gray canvas with blue/pink accents) was applied to ensure color-blind accessibility.

Clutter Reduction: The data-to-ink ratio was maximized by removing unnecessary background gridlines and using direct data labels instead of redundant Y-axes.

Visual Selection: Chart types were deliberately chosen based on data shape, utilizing donut charts for simple proportions and line charts for time-series trends.

Explanation of How Users Interact with the Dashboard

Global Slicers: A persistent left-hand slicer panel allows administrators to instantly filter the entire page by specific admission years or facility locations.

Cross-Filtering: Every visual is fully interactive; clicking a specific bar or pie slice dynamically filters all other charts on the page to highlight that data point.

Hierarchical Drill-Downs: The column charts feature drill-down capabilities, allowing users to dive seamlessly from high-level facility data into specific patient comorbidities.

Part F: Analytical Insights and Business Recommendations

Analytical Insights

By cross-filtering the dashboard visuals and utilizing the AI decomposition features, I identified the following critical patterns and risks within the hospital network:

Geographical Capacity Bottleneck at Facility E: While the network-wide Average Length of Stay (LOS) is performing well at 4.00 days, the system is being heavily dragged down by Facility E. As seen on Page 1 and Page 2, Facility E averages an alarming 5.16 days per encounter, significantly higher than highly efficient locations like Facilities A and B (~3.2 days).

Compounding Effect of Patient Risk Tiers: The Decomposition Tree shows that patient risk does not scale linearly; it compounds based on location. A "High Risk" patient at Facility E stays for an average of 7.92 days nearly double the network average. This indicates a severe struggle in managing complex, multi readmission patients at this specific location.

Clinical Drivers of Extended Stays: The Key Influencers AI visual identified a direct correlation between admission lab results and operational strain. Specifically, a drop in patient glucose levels drives the Average LOS up by 1.1 days. This proves that metabolic instability is a primary clinical indicator of a prolonged, costly hospitalization.

Alarming Readmission Burden: Despite the donut chart showing that the majority of the patient population (55.03%) has "No Prior Readmission", the network is still battling a massive 45% overall readmission rate. This suggests a systemic issue with premature discharges or failing post-hospital care regimens across the network.

Significant Demographic Skew: The patient demographic is heavily skewed towards female patients (57.64% Female vs. 42.36% Male). Furthermore, the facility-level cross-filtering shows that female patients make up the overwhelming volume of admissions at the highest-burden locations, presenting a clear demographic focal point for community health initiatives.

Actionable Business Recommendations

Based on the data insights, hospital administration should implement the following strategic initiatives:

Immediate Resource Reallocation to Facility E:

Hospital executives must urgently audit the discharge procedures and staffing ratios at Facility E. Because Facilities A and B are operating highly efficiently, the network should temporarily route incoming non emergency patients away from Facility E, and consider reallocating specialized medical staff from A and B to help clear Facility E's backlog of extended stay patients.

Implement a Predictive Metabolic Protocol:

Because dropping glucose levels add over a full day to a patient's stay, the clinical directors must establish a strict, standardized metabolic monitoring protocol at the triage stage. By aggressively stabilizing glucose levels upon initial admission before routing patients to general wards, the hospital can proactively prevent an estimated 1.1 days of extended care per affected patient, saving millions in operational costs.

Establish a High-Risk Post-Discharge Program:

To combat the severe 45% readmission rate and the nearly 8-day stays of high-risk cohorts, the network must invest in preventative post-discharge care. The hospital should launch a targeted Telehealth and Visiting Nurse program specifically for the 20K patients categorized in the "High Risk" tier. Monitoring these patients in their homes will prevent their conditions from deteriorating to the point of requiring another massive, multi-day hospital admission.